

'Balance The Books' With FreeCAD

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Designers struggle to find the right mechanical and electronic design and simulation software. Innumerable meetings and brainstorming over alternatives eat into their time and energy. Worse, they end up buying the software, paying a hefty amount for it.

Thankfully, a saviour has emerged. FreeCAD is an open source mechanical CAD freeware that is as good as its paid counterparts. Its latest version, VER 0.16, is out in the market and ready to ease the work of designers.

A no-cost, easy-to-use tool

FreeCAD is a user-friendly tool with features of design, simulation and strength calculations for the product. Users need no prior training to be able to use it. After design, a mesh

frame is created, which is followed by simulation (Fig. 1).

Consider the case when a designer is making metallic housing for an LED street-light. The designer needs to understand areas that would be under more mechanical stress or require more strength. He also needs to understand thermal management in the housing. Also, there will be calculations related to mechanical functions. The FreeCAD software comes handy here: It would first work on the design workbench, then mesh for strengthening the design, and finally simulation.

How it helps

The latest version of FreeCAD, 0.16, has some additional features:

Expression Support. FreeCAD helps make models using Excel

spreadsheet format, just like in CATIA. Called Expression Support, this helps designers to get the precise location of coordinates. FreeCAD works on the concept of workbench.

Sketcher Solver upgrade. Sketcher Solver was introduced in the previous version (0.15), but in the new version (0.16) it makes sketch editing



Fig. 2: Sketcher Solver (Image courtesy: www.opennet.ru)

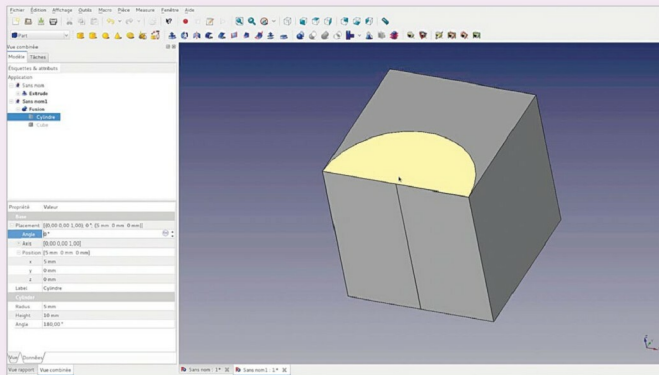


Fig. 1: 3D view in FreeCAD (Image courtesy: www.freecadweb.org)